

Vertical Monotonicity of de Branges–Rovnyak Kernel Norms

A full answer to Question 6.6 of arXiv:0802.0789

Autonomous proof packet for `fa_banach_001`

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Status. Candidate full result, likely valid, pending human review. The result answers the exact monotonicity question affirmatively and in the larger range $1 < q < \infty$.

1 Source question

Let b be in the closed unit ball of $H^\infty(\mathbb{C}_+)$. Its de Branges–Rovnyak kernel, viewed through its boundary values, is

$$k_z^b(t) = \frac{1 - \overline{b(z)}b(t)}{t - \bar{z}}, \quad z \in \mathbb{C}_+, \quad t \in \mathbb{R}.$$

Baranov, Fricain, and Mashreghi proved for inner b that, for $q > 1$, there is a constant depending only on q such that

$$\|k_{x+iy_1}^b\|_q \leq C(q) \|k_{x+iy_2}^b\|_q \quad (x \in \mathbb{R}, \quad 0 \leq y_2 \leq y_1), \quad (6.9)$$

and ask in Question 6.6 whether this remains true for a general b [1, Question 6.6, PDF pp. 27–28].

[6.2](#) we used the density in $\mathcal{H}(b)$ of the functions continuous on all squares S_k .

Question 6.4. Let b be in the unit ball of $H^\infty(\mathbb{C}_+)$. Is it true that the set of functions f in $\mathcal{H}(b)$, continuous on $\overline{\mathbb{C}_+}$, is dense in $\mathcal{H}(b)$?

Remark 6.5. In the inner case, in Theorem [6.2](#) the assumption [\(6.3\)](#) can be replaced by the weaker assumption (only for the lower side of the square)

$$(6.8) \quad \sup_{k \geq 1} |I_k| \int_{I_k} w_p^{-2}(u) |du| < \infty.$$

It was noticed in [\[6 Corollary 4.7\]](#) that in the inner case, for $q > 1$, there exists $C = C(q) > 0$ such that, for any $x \in \mathbb{R}$ and $0 \leq y_2 \leq y_1$, we have

$$(6.9) \quad \|k_{x+iy_1}^b\|_q \leq C(q) \|k_{x+iy_2}^b\|_q.$$

Source PDF page 27: the inequality referred to as (6.9).

Thus, it follows from (6.9) that if the sequence $\{S_k\}$ satisfies (6.8), then it also satisfies (6.3).

Question 6.6. Does the monotonicity property (6.9) of the norms of the reproducing kernels along the rays parallel to imaginary axis remains true for a general b ? (It is true for $q = 2$, but this is not the interesting case for us.)

Proof. of Theorem 6.7. (a) Consider the open set $E = \mathbb{R} \setminus \text{Clos } \widetilde{\Omega}(b, \varepsilon)$. If $E = \emptyset$, then μ

Source PDF page 28: complete statement of Question 6.6.

2 Idea of the proof

The kernel at the higher point is exactly the lower kernel multiplied by a boundary contraction, plus one ordinary Cauchy kernel. The coefficient of that error is $b(z_1) - b(z_2)$. The elementary geometry of the unit disk bounds this coefficient by $|1 - \overline{b(z_2)}b(z_1)|$. But the latter quantity is precisely the value of the lower kernel at z_1 , times $|z_1 - \overline{z_2}|$. A single H^q point-evaluation estimate therefore absorbs the entire error into the lower kernel norm. No innerness or boundary modulus-one property of b is used.

3 The theorem

For $1 < s < \infty$, put

$$A_s := \left(\int_{\mathbb{R}} (1 + u^2)^{-s/2} du \right)^{1/s}.$$

Theorem 1 (vertical kernel monotonicity). *Let $b \in H^\infty(\mathbb{C}_+)$ satisfy $\|b\|_\infty \leq 1$, let $1 < q < \infty$, and let $q' = q/(q-1)$. For every $x \in \mathbb{R}$ and $0 < y_2 \leq y_1$,*

$$\|k_{x+iy_1}^b\|_{L^q(\mathbb{R})} \leq \left(1 + \frac{A_q A_{q'}}{\pi} \right) \|k_{x+iy_2}^b\|_{L^q(\mathbb{R})}. \quad (1)$$

The same conclusion holds for $y_2 = 0$ whenever the boundary kernel k_x^b exists as an $H^q(\mathbb{C}_+)$ function. If its extended L^q norm is infinite, the inequality is automatic.

Proof. Write

$$z_j = x + iy_j, \quad b_j = b(z_j), \quad k_j = k_{z_j}^b \quad (j = 1, 2).$$

For real t , direct subtraction of the two kernel numerators gives

$$k_1(t) = \frac{t - \overline{z_2}}{t - \overline{z_1}} k_2(t) + \frac{(\overline{b_2} - \overline{b_1})b(t)}{t - \overline{z_1}}. \quad (2)$$

Because $0 \leq y_2 \leq y_1$,

$$\left| \frac{t - \overline{z_2}}{t - \overline{z_1}} \right|^2 = \frac{(t - x)^2 + y_2^2}{(t - x)^2 + y_1^2} \leq 1,$$

and $|b(t)| \leq 1$ almost everywhere. Hence

$$\|k_1\|_q \leq \|k_2\|_q + |b_1 - b_2| \left\| \frac{1}{t - \bar{z}_1} \right\|_q = \|k_2\|_q + A_q |b_1 - b_2| y_1^{1/q-1}. \quad (3)$$

We next absorb the second term. The elementary identity

$$|1 - \bar{a}w|^2 - |w - a|^2 = (1 - |a|^2)(1 - |w|^2) \geq 0$$

for a, w in the closed unit disk gives

$$|b_1 - b_2| \leq |1 - \bar{b}_2 b_1|. \quad (4)$$

For completeness, the standard H^q point-evaluation estimate follows immediately from the Cauchy formula and Hölder's inequality: if $f \in H^q(\mathbb{C}_+)$ and $z = x + iy$, then

$$|f(z)| \leq \frac{1}{2\pi} \|f\|_q \left\| \frac{1}{t - z} \right\|_{q'} = \frac{A_{q'}}{2\pi} y^{-1/q} \|f\|_q. \quad (5)$$

Apply this to $f = k_2$ at z_1 . Since

$$k_2(z_1) = \frac{1 - \bar{b}_2 b_1}{z_1 - \bar{z}_2}, \quad |z_1 - \bar{z}_2| = y_1 + y_2 \leq 2y_1,$$

estimate (5) yields

$$|1 - \bar{b}_2 b_1| y_1^{1/q-1} \leq \frac{A_{q'}}{\pi} \|k_2\|_q. \quad (6)$$

Combining (3), (4), and (6) proves (1).

When $y_2 = 0$ and $k_x^b \in H^q$, the same proof is literal: set $z_2 = x$ and let $b_2 = b(x)$ in the boundary-kernel definition. The multiplier in (2) still has modulus at most one, and now $|z_1 - \bar{z}_2| = y_1$. This proves the boundary case as well. $\square$

Corollary 1. *Question 6.6 of the source has an affirmative answer. In particular, the constant is independent of b , x , y_1 , and y_2 , and the conclusion holds throughout the source's range $q \geq 2$.*

4 Stress tests and scope

The proof was checked against the following edge cases. For a unimodular constant b , all kernels vanish. For a constant $|b| < 1$, both sides have the expected scaling $y^{1/q-1}$. At $q = 2$, $A_2 = \sqrt{\pi}$, so the displayed constant is 2, consistent with the known case though not asserted to be sharp. The proof remains uniform as $y_2 \downarrow 0$ whenever a finite H^q boundary kernel is present.

No computational evidence is used. Question 6.6 is solved in full, but this packet makes no claim about the distinct boundary-integrability Question 3.3 in the source. The source's Question 6.4 was answered later by Aleman–Malman; that separate literature fact is not used in this proof.

5 Novelty check and human review

A bounded run-index and web/arXiv search on 17 August 2026 used arXiv id 0802.0789, the exact wording of Question 6.6, and combinations of “de Branges–Rovnyak”, “vertical rays”, “reproducing kernel”, “ L^q norm”, and “monotonicity”. The search found the source and adjacent kernel literature but no later statement of the general- b inequality or this two-kernel proof. Novelty confidence is moderate rather than conclusive.

Human review is recommended. The decisive checks are the kernel identity (2), the normalization in the Cauchy estimate (5), and the source's convention for a boundary kernel when $y_2 = 0$.

References

- [1] A. Baranov, E. Fricain, and J. Mashreghi, *Weighted norm inequalities for de Branges–Rovnyak spaces and their applications*, arXiv:0802.0789, 2008.